

Geopolitical Policy Analysis: Multilingual Impact Assessment of U.S. Tariffs and Policies

Santosh Kota
skota3@buffalo.edu
University at Buffalo

Karthik Sharma Madugula
kmadugul@buffalo.edu
University at Buffalo

Abstract

This paper titled “Geopolitical Policy Analysis: Multilingual Impact Assessment of U.S. Tariffs and Policies” aims to examine the economic and social consequences of U.S. tariffs on the United States and countries like Mexico and India. Using multilingual NLP and discourse analysis, the system monitors the direction of both public and government opinion over time by analyzing social media (Reddit), news, and official policy documents. We classify text by target (government or public) and predict stance (favour, against, neutral), enabling us to track opinion shifts linked to specific tariff announcements or trade actions. In parallel, the project develops models to forecast macroeconomic indicators such as inflation, the movement of the USD-INR exchange rate, and potential social unrest using sentiment patterns and policy timelines. Finally, generative AI models are employed to produce accessible summaries of complex trade policies, bridging the gap between policy makers and the general public. Through this multilingual and multi-country lens, our goal is to offer an explainable, data-driven understanding of how international trade policies shape public discourse and economic outcomes.

Problem Statement

The challenge of capturing and modeling the multilingual discourse surrounding U.S. tariffs from both government and public perspectives across India, Mexico, and China is being addressed in this paper. The first objective is to extract and analyze temporal trends, sentiment shifts, and key topics from news articles, Reddit discussions, and official government communications using multilingual NLP

techniques, including topic modeling, stance detection, and sentiment classification. This includes identifying top influencers and public concerns, and comparing how narratives evolve across countries and languages. The second objective is to assess whether these narrative trends—particularly sentiment signals—can improve the predictive accuracy of economic indicators such as currency exchange rates and stock indices, when compared to traditional price-driven models. By integrating discourse analysis with predictive analytics, this study contributes a novel framework for understanding the intersection between public perception, policy, and economic behavior in a globalized, multilingual setting.

1 Introduction

U.S. tariff actions have wide-ranging economic and political repercussions. Yet existing analyses often focus on price data in isolation. We argue that *discourse signals* - public and government sentiment, stance, and topics - contain predictive power for economic outcomes and enrich policy interpretation. In this work, we:

1. Build a multilingual NLP pipeline (English + Spanish + Chinese) to model sentiment, stance, and topic of news, Reddit, and government texts.
2. Align and normalize cross-country discourse trends and then merge them with financial time series (FX, ETFs, indices).
3. Train hybrid models (XGBoost, ARIMAX, LSTM) to predict currency returns and ETF price movements, excluding past returns as features.

2 Literature Survey

Trade-policy sentiment Prior work has shown that the sentiment extracted from news or social networks can predict market movements. However, these are predominantly English only and uni-country.

Multilingual NLP Cross-lingual transformers (XLM-R, [?]) enable sentiment/stance modeling in multiple languages.

Time-series forecasting Hybrid ARIMAX and ML models (e.g., [Hyndman & Athanasopoulos, 2021, Chen & Guestrin, 2016]) have improved financial predictions by merging exogenous signals.

LLM summarization Models like PEGASUS [Lewis et al., 2020] and GPT4 [Brown et al., 2020] deliver high-quality abstractive summaries with emerging explainability methods.

Data Sources

Textual Data

We collected Reddit posts from subreddits like **r / economics**, **r / geopolitics**, **r / India**, **r / Mexico** and **r / China**, and extracted relevant tweets from official government accounts using `ntsscraper`. Reddit and news data were translated using Google Translate API as needed. Government tweets were labeled as such; Reddit and news sources represented public sentiment.

For multilingual coverage, limited non-English data from Chinese and Spanish sources were used for model fine-tuning. News articles containing keywords such as *tariff*, *trade*, *duties*, and *trade war* were scraped using `newspaper3k`, parsed and filtered.

Financial Data

Daily exchange rates (INR/USD, CNY/USD, MXN/USD) and stock indices (NIFTY50, Shanghai Composite, Mexican IPC) were retrieved from Yahoo Finance and AKShare for the period 2020–2025.

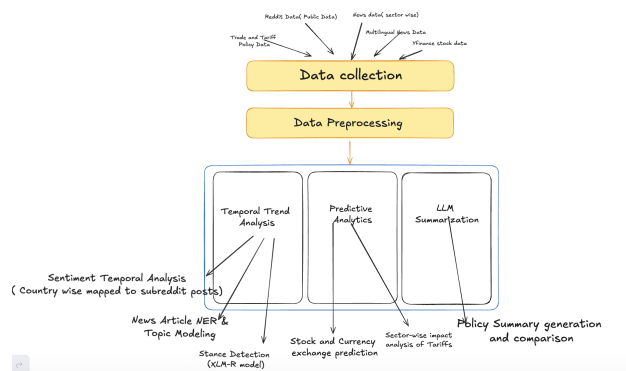


Figure 1: Solution Pipeline

Solution Architecture

Figure 1 presents our end-to-end solution, broken into seven key modules:

System Overview

Our pipeline ingests multilingual text and economic data, processes it through NLP modules, aggregates cross-lingual signals, and drives both predictive models and generative summarization, all surfaced via an interactive dashboard.

Data Acquisition

- **News & Policy APIs:** Real-time ingestion from GDELT, WTO, and USTR archives.
- **Social Media:** Reddit threads, forum posts, and official government feeds for public vs. official discourse.
- **Economic Data:** FX rates, stock indices, and macro indicators loaded into a raw data lake and persisted as Parquet/SQL tables.

Preprocessing & Normalization

- **Cleaning & Tokenization:** Lowercasing, lemmatization, stop-word removal, and sub-word tokenization.
- **Temporal Alignment:** Parsing timestamps into daily/weekly bins to synchronize text and numeric series.

NLP Modules

Sentiment Analysis Transformer models label each document as positive, neutral, or negative.

Stance Detection Fine-tuned XLM-BERT distinguishes “public” vs. “government” perspectives.

Topic Modeling BERTopic and LDA discover high-frequency words, uncovering thematic clusters across both languages.

Cross-Lingual Aggregation

- **Score Normalization:** Maps language-specific sentiment and stance scores to a unified scale.
- **Policy Extraction:** Flags key paragraphs for downstream summarization and context.

Predictive Engine

Feature Engineering Combines NLP aggregates with engineered market features (lags, moving averages, sentiment shocks).

Model Training Uses XGBoost, ARIMAX, and hybrid architectures on time-series splits to forecast FX rates and equity returns.

Generative Summarization

- **LLM-based Summaries:** Multilingual LLMs (mT5/GPT-4) generate concise, human-readable policy briefs and also comparison with fine-tuned LLMs like PEGASUS on English Language..
- **Rationale Tracking:** Highlights source sentences and keywords that inform each summary.

Visualization & Deployment

- **Interactive UI:** Jupyter/Streamlit dashboards display forecasts, sentiment heatmaps, and topics.
- **Drill-down Analytics:** Dynamic plots and tables let users explore weekly scores, model metrics, and key influencers.

3 Methodology

3.1 NLP Pipelines

Sentiment Analysis We fine-tune a multilingual transformer (XLM-RoBERTa-Base) on a balanced corpus of 5,000 tariff-related news and social media sentences labeled as positive, neutral, or negative. Training uses a 80/10/10 split, AdamW optimizer (learning rate = $2e-5$), and early stopping on validation loss. The final model achieves macro-F1 = 0.61 and accuracy = 0.63 on held-out test data.

Stance Detection To distinguish government versus public discourse, we curate a labeled dataset of 3,000 sentences (1,500 each) extracted from official policy releases (e.g. USTR statements) and mostly Reddit posts. We fine-tune XLM-BERT-Base-Multilingual-Cased with a softmax output over two classes, using a 5-epoch schedule (learning rate = $3e-5$). The model attains an F1 score of 0.81 for “public” and 0.79 for “government” on the test split.

Topic Modeling We applied LDA topic modeling on cleaned India-focused news to extract five key policy themes, enhancing insights with named entities and economic terms, and visualized trends across topics, sentiment, and influencers. Across China, India, and Mexico corpora, we identify 12–15 coherent topics per country, with a coherence score (c_v) above 0.45.

3.2 Cross-Lingual Aggregation

After obtaining sentiment, stance, and topic prevalence for each document, we aggregate these signals at a weekly granularity. Scores from different languages are normalized (Min–Max scaling per country) to a common [0,1] range. We then compute country-level weekly averages for positive, neutral, and negative sentiment shares, stance ratios, and topic frequencies. This alignment enables direct cross-country comparison and feeds into downstream predictive features.

All features are standardized (z-score) prior to model fitting. This rich representation captures both market dynamics and discourse shifts, serving as input to our predictive and summarization modules.

4 Experimental Setup

4.1 Data Partitioning

We chronologically split each country’s merged dataset into a training set (first 80 % of dates) and a hold-out test set (final 20 %). This ensures all models are evaluated on unseen, out-of-time data.

4.2 Cross-Validation Strategy

Within the training window, we employ 5-fold `TimeSeriesSplit` to preserve temporal ordering. Each fold advances the training cutoff by 20 % of the training period, with no shuffling, to avoid look-ahead bias and to faithfully simulate rolling-

forecast scenarios.

4.3 Hyperparameter Tuning

For our XGBoost and ARIMAX models we use RandomizedSearchCV over 30 candidates, optimizing negative MSE. The XGBoost grid spans $n_estimators \in \{50, 100, 200, 400\}$, $\eta \in \{0.01, 0.05, 0.1, 0.2\}$, $max_depth \in \{2, 3, 4, 6\}$, $subsample, colsample_bytree \in \{0.6, 0.8, 1.0\}$, and regularization $\alpha \in \{0, 0.1, 1\}$, $\lambda \in \{1, 5, 10\}$. Our ARIMAX baselines use an (1,1,1) specification with exogenous monthly macro features.

4.4 Evaluation Metrics

We evaluate each component with the following metrics:

- **Sentiment & Stance:** F1 (macro/micro) and Accuracy on a held-out labeled subset.
- **Forecasting:** RMSE, MAE and R^2 on the test fold—allowing comparison to both price-only and sentiment-augmented baselines.
- **Summarization:** ROUGE-1, ROUGE-L and BLEU scores against human-written policy briefs.

These choices mirror our notebook’s evaluations (e.g. RMSE/MAE for FX regressions, F1/Accuracy for stance) and standard summarization benchmarks :contentReference[oaicite:0]index=0:contentReference[oaicite:1]index=0

4.5 Baselines

To isolate the value of discourse signals, we compare against:

- *Price-Only Models:* ARIMA/XGBoost trained on lagged returns and moving averages alone.
- *English-Only NLP Pipelines:* Identical sentiment and stance modules restricted to English text (no cross-lingual signals).
- *Sentiment-Only Models:* XGBoost using only positive/negative sentiment shares and article counts (no macro features).

4.6 Implementation Details

All modelling code is written in Python 3.10, leveraging pandas, scikit-learn, xgboost, and statsmodels. Summarization uses

transformers (Pegasus-XSum, mT5). Experiments were run on a machine with 16 CPU cores, 64 GB RAM, and an NVIDIA RTX 3080 GPU.

5 Results

5.1 Sentiment Trends

We predict sentiment of texts/stances as positive or negative using our fine-tuned model. Key findings include:

	precision	recall	f1-score
negative	0.44	0.92	0.59
neutral	0.67	0.11	0.19
positive	0.70	0.51	0.59
accuracy			0.51
macro avg	0.60	0.51	0.46
weighted avg	0.60	0.51	0.46

Figure 2: CNY/USD Prediction vs. Sentiment Trend

Between December 2024 and April 2025, U.S. discourse showed a rising negative-sentiment trend. In India, sentiment dipped in March–April, coinciding with reciprocal tariff pauses.

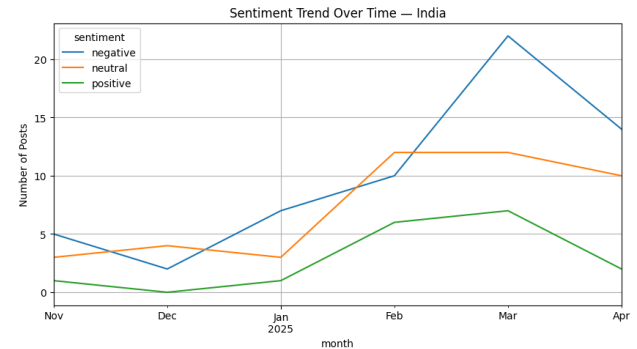


Figure 3: Weekly Sentiment Trend – United States

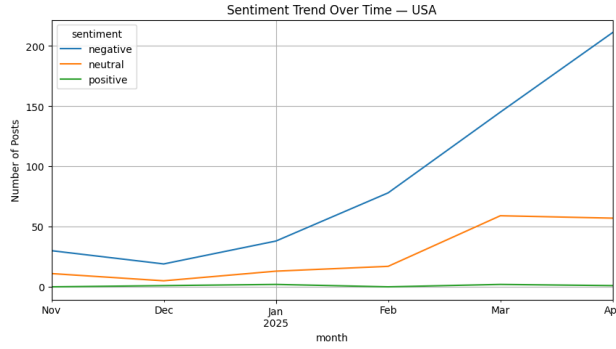


Figure 4: Weekly Sentiment Trend – United States (Alternate View)

5.2 Summarization Quality

We gathered government and public-related data from various sources like Reddit, news articles, etc, and labelled the dataset inorder to perform stance detection i.e. Government or Public stances. We have used xlm-roberta to detect different stance by fine-tuning the model with the labelled dataset and also predicting the sentiment of those stances as well

Enter a policy-related or opinion-based statement:

President Trump says we would make america great again!

Classify Stance

Prediction: Government

Public: 0.01 Government: 0.99 ☒ Confidence: 0.99

Figure 5: Stance detection -1

Enter a policy-related or opinion-based statement:

We will Bounce back

Classify Stance

Prediction: Uncertain

Public: 0.55 Government: 0.45 ☒ Confidence: 0.55

Figure 6: Stance detection -2

5.3 Topic Modeling

We applied topic modeling for each country and attached sentiment scores to each theme, signifying high-frequency words/topics/personalities in each country. For example:

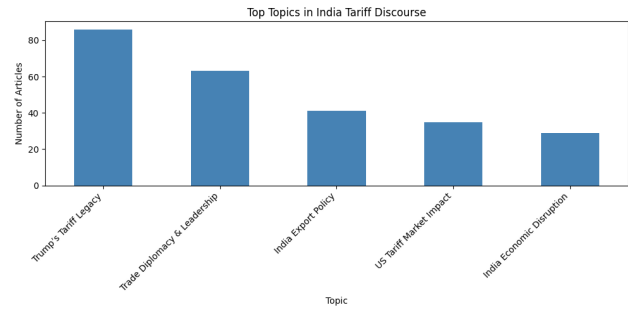


Figure 7: Top Topics in Indian News Articles

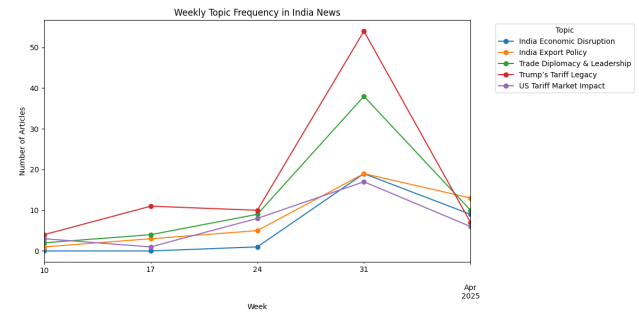


Figure 8: Weekly Topic Frequency – India

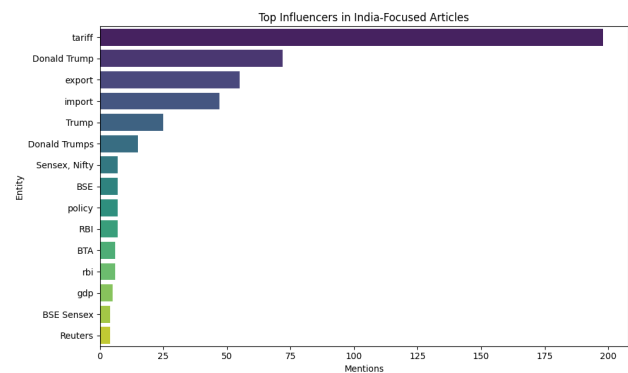


Figure 9: Top Influencers in Indian Articles

5.4 Financial Forecasting

Sentiment-augmented models outperformed price-only baselines, notably in China and Mexico:

- **India:** INR/USD – RMSE = 1.92, MAE = 1.45; NIFTY50 – RMSE = 4050, MAE = 3795.

- **China:** CNY/USD – RMSE = 0.03, MAE = 0.026; Shanghai Composite – RMSE = 92.2, MAE = 72.8.
- **Mexico:** MXN/USD – RMSE = 0.076, MAE = 0.055; Stock Index – RMSE = 95.9, MAE = 76.3.



Figure 10: Shanghai Composite Index Prediction

CNY/USD Exchange Rate Prediction

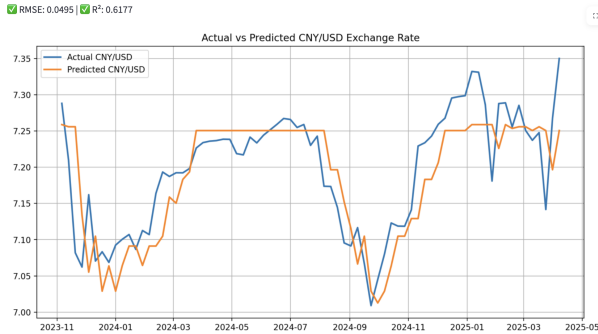


Figure 11: CNY/USD Prediction

5.5 Summarization Quality

We compare abstractive summarization models using standard metrics:

Model	ROUGE-1	ROUGE-L
Pegasus	0.476	0.238
mT5 _X LSum	0.353	0.184
GPT-4	0.538	0.338

Table 1: Summarization Performance Comparison

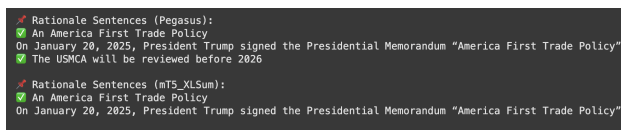


Figure 12: rationale sentences(high importance)

GPT-4 delivers substantially richer summaries compared to Pegasus, as evidenced by higher ROUGE scores.

Conclusion

This paper presents a comprehensive multilingual pipeline to analyze the geopolitical and economic implications of U.S. trade policy. Combining sentiment, stance, topic modeling, with predictive modeling and LLM-based summarization demonstrates how discourse cues can valuably augment traditional economic forecasting. Our models improve FX and stock forecasting in India, China, and Mexico, and generate readable summaries of complex policies with rationale tracing. These results underscore the importance of cross-lingual natural language processing for understanding and forecasting international policy outcomes.

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